

- 301.** Let $\alpha > 0$, be the smallest number such that the expansion of $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ has term $\beta x^{-\alpha}$, $\beta \in \mathbb{N}$. Then α is equal to _____.

माना $\alpha > 0$ न्यूनतम संख्या है, जिसके लिए $\left(x^{\frac{2}{3}} + \frac{2}{x^3}\right)^{30}$ के प्रसार के एक पद में $\beta x^{-\alpha}$ तो α बराबर है _____, $\beta \in \mathbb{N}$

Ans. 2

Sol. $\alpha > 0$, $\left(x^{2/3} + \frac{2}{x^3}\right)^{30}$

$$T_{r+1} = {}^{30}C_r \left(x^{2/3}\right)^{30-r} \left(\frac{2}{x^3}\right)^r$$

$$= {}^{30}C_r x^{20 - \frac{2r}{3} - 3r}$$

$$= {}^{30}C_r (2)^r x^{\frac{60-11r}{3}} \quad 0 \leq r \leq 30$$

For $\beta x^{-\alpha}$, $\alpha > 0$ the smallest value of α is 2 (for $r = 6$)

$$T_7 = {}^{30}C_6 (2)^6 x^{-2}$$

$$\alpha = 2$$

- 302.** The remainder on dividing 5^{99} by 11 is _____.

5^{99} को 11 से विभाजित करने पर शेषफल _____ है।

Ans. 9

Sol. $5^{99} = 5^4 \cdot 5^{95}$

$$= 625 [5^5]^{19}$$

$$= 625 [3125]^{19}$$

$$= 625 [2124 + 1]^{19}$$

$$= 625 [11k \times 19 + 1]$$

$$= 625 \times 11k \times 19 + 625$$

$$= 11k_1 + 616 + 9$$

$$= 11(k_2) + 9$$

$$\text{Remainder} = 9$$

- 303.** The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____.

$19^{200} + 23^{200}$ को 49 से विभाजित करने पर शेषफल _____ है।

Ans. 29

Sol. $(21 - 2)^{200} + (21 + 2)^{200} = 49\lambda + 2^{201}$
 $2^{201} = 8^{67} = (7 + 1)^{67} = 49\lambda + 7 \times 67 + 1$
 $= 49\lambda + 470$
 $= 49(\lambda + 9) + 29$

So, remainder = 29

304. Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is _____.

माना $2^{(x-2)\log_2 3}$ की बढ़ती घातों में $\left(\sqrt{2^{\log_2(10-3^x)}} + \sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$ के द्विपद प्रसार में छठा पद 21 है। यदि इस प्रसार में दूसरा, तीसरा तथा चौथा द्विपद गुणांक एक A.P. के क्रमशः पहला, तीसरा तथा पाँचवा पद है, तो x के सभी संभव मानों के वर्गों का योग है।

Ans. 4

Sol. $T_6 = {}^m C_5 (10 - 3^x)^{\frac{m-5}{2}} (3^{x-2}) = 21 \quad \dots(1)$

${}^m C_1, {}^m C_2, {}^m C_3$ are in A.P.

$$2 \cdot {}^m C_2 = {}^m C_1 + {}^m C_3$$

Solving for m , we get

$$m = 2 \text{ (rejected), } 7$$

Put in equation (1)

$$21 \cdot (10 - 3^x) \frac{3^x}{9} = 21$$

$$3^x = 3^0, 3^2$$

$$x = 0, 2$$

Sum of all the squares of all possible values of $x = 4$

305. Let α be the constant term in the binomial expansion of $\left(\sqrt{x} - \frac{6}{x^2}\right)^n$, $n \leq 15$. If the sum of the coefficients of the remaining terms in the expansion is 649 and the coefficient of x^{-n} is $\lambda\alpha$, then λ is equal to _____.

माना $\left(\sqrt{x} - \frac{6}{x^{\frac{3}{2}}}\right)^n$, $n \leq 15$ के द्विपद प्रसार में अचर पद α है। यदि इस प्रसार में शेष पदों के गुणांकों का योग 649 है तथा

x^{-n} का गुणांक $\lambda\alpha$ है, तो λ बराबर है _____

Ans. 36

Sol. $T_{k+1} = {}^nC_k (x)^{\frac{n-k}{2}} (-6)^k (x)^{\frac{-3}{2}k}$

$$\frac{n-k}{2} - \frac{3}{2}k = 0$$

$$n - 4k = 0$$

$$(-5)^n - \left({}^nC_{\frac{n}{4}} (-6)^{\frac{n}{4}}\right) = 649$$

By observation ($625 + 24 = 649$), we get $n = 4$

$\therefore n = 4$ & $k = 1$

Required is coefficient of x^{-4} is $\left(\sqrt{4} - \frac{6}{x^{\frac{3}{2}}}\right)^4$

$${}^4C_1 (-6)^3$$

By calculating we will get $\lambda = 36$

306. The remainder, when 7^{103} is divided by 17, is _____.

7^{103} को 17 से विभाजित करने पर शेषफल है _____

Ans. 12

Sol. 7^{103} divided by 17

$$7 \equiv 7 \pmod{17}$$

$$7^2 \equiv -2 \pmod{17}$$

$$7^6 \equiv -8 \pmod{17}$$

$$7^8 \equiv -1 \pmod{17}$$

$$7^{16} \equiv 1 \pmod{17}$$

$$7^{103} \equiv 12 \pmod{17}$$

$\therefore$ Remainder = 12

307. The number of elements in the set $\{n \in \mathbb{N} : 10 \leq n \leq 100 \text{ and } 3^n - 3 \text{ is a multiple of } 7\}$ is _____.

समुच्चय में अवयवों की संख्या होगी $\{n \in \mathbb{N} : 10 \leq n \leq 100 \text{ तथा } 3^n - 3, 7 \text{ का गुणक}\}$ _____

Ans. 15

Sol. $n \in [10, 100]$
 $3^n - 3$ is multiple of 7
 $3^n = 7\lambda + 3$
 $n = 1, 7, 13, 20, \dots, 97$
 Number of possible values of $n = 15$

Complex Number

308. Let z be a complex number such that $|z| = 1$. If $\frac{2+k^2z}{k+\bar{z}} = kz, k \in \mathbf{R}$ then the maximum distance of $k + ik^2$ from the circle $|z - (1 + 2i)| = 1$ is :

माना z एक सम्मिश्र संख्या है जिसके लिए $|z| = 1$ है। यदि $\frac{2+k^2z}{k+\bar{z}} = kz, k \in \mathbf{R}$ है, तो वृत्त $|z - (1 + 2i)| = 1$ से $k + ik^2$ की

अधिकतम दूरी है :

- (A) 3 (B) $\sqrt{3} + 1$ (C) $\sqrt{5} + 1$ (D) 2

Ans. C

Sol. $\frac{2+k^2z}{k+\bar{z}} = kz$
 $|z|^2 k = 2$
 $k = 2$
 point $p(2, 4)$; center $(1, 2)$
 distance from circle
 $(x-1)^2 + (y-2)^2 = 1$ is max.
 if $(OP + r) = \sqrt{1+4} + 1 = \sqrt{5} + 1$

309. Let $z \in \mathbf{C}$ be such that $\frac{z^2 + 3i}{z - 2 + i} = 2 + 3i$. Then the sum of all possible values of z^2 is :

माना $z \in \mathbf{C}$ के लिए $\frac{z^2 + 3i}{z - 2 + i} = 2 + 3i$ है। तो z^2 के सभी संभव मानों का योग है :

- (A) $-19 + 2i$ (B) $-19 - 2i$ (C) $19 - 2i$ (D) $19 + 2i$

Ans. B

Sol. $z^2 + 3i = z(2 + 3i) - 7 - 4i$
 $z^2 - z(2 + 3i) + 7 + 7i = 0$
 z_1
 z_2
 $z_1^2 + z_2^2 = (z_1 + z_2)^2 - 2z_1 z_2$
 $= -19 - 2i$

310. If $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of an equilateral triangle, whose centroid is z_0 , then $\sum_{k=1}^3 (z_k - z_0)^2$ is equal to :

माना एक समबाहु त्रिभुज के शीर्ष $z_1, z_2, z_3 \in \mathbb{C}$ हैं। यदि इसका केन्द्रक z_0 है, तो $\sum_{k=1}^3 (z_k - z_0)^2$ बराबर है :

- (A) $-i$ (B) 1 (C) 0 (D) i

Ans. C

Sol. $z_1 + z_2 + z_3 = 3z_0$

$$(z_1 + z_2 + z_3)^2 = 9z_0^2$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 + 2(z_1 z_2 + z_2 z_3 + z_3 z_1) = 9z_0^2$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 = 9z_0^2 - 2(z_1 z_2 + z_2 z_3 + z_3 z_1)$$

$$\sum_{k=1}^3 (z_k - z_0)^2 = (z_1 - z_0)^2 + (z_2 - z_0)^2 + (z_3 - z_0)^2$$

$$= z_1^2 + z_2^2 + z_3^2 + 3z_0^2 - 2(z_1 + z_2 + z_3)z_0$$

$$= 6z_0^2 - 6z_0^2$$

$$= 0$$

311. Let the product of $\omega_1 = (8 + i) \sin \theta + (7 + 4i) \cos \theta$ and $\omega_2 = (1 + 8i) \sin \theta + (4 + 7i) \cos \theta$ be $\alpha + i\beta, i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then $p + q$ is equal to :

माना $\omega_1 = (8 + i) \sin \theta + (7 + 4i) \cos \theta$ और $\omega_2 = (1 + 8i) \sin \theta + (4 + 7i) \cos \theta, i = \sqrt{-1}$, का गुणनफल $\alpha + i\beta$ है। माना $\alpha + \beta$ के अधिकतम और न्यूनतम मान क्रमशः p और q हैं। तो $p + q$ बराबर है :

- (A) 160 (B) 150 (C) 140 (D) 130

Ans. D

Sol. $w_1 w_2 = (8 + i)(1 + 8i) \sin^2 \theta + (8 + i)(4 + 7i) \sin \theta \cos \theta + (7 + 4i)(1 + 8i) \sin \theta \cos \theta + (7 + 4i)(4 + 7i) \cos^2 \theta$

$$w_1 w_2 = (65i) \sin^2 \theta + (120i) \sin \theta \cos \theta + (65i) \cos^2 \theta$$

$$= 65i + 60i (\sin 2\theta)$$

$$\alpha = 0, \beta = 65 + 60 \sin 2\theta$$

$$p = 125, q = 5$$

$$p + q = 130$$

312. Among the statements

(S1) : The set $\{z \in \mathbb{C} - \{-i\} : |z| = 1 \text{ and } \frac{z-i}{z+i} \text{ is purely real}\}$ contains exactly two elements, and

(S2) : The set $\{z \in \mathbb{C} - \{-1\} : |z| = 1 \text{ and } \frac{z-1}{z+1} \text{ is purely imaginary}\}$ contains infinitely many elements.

(A) both are incorrect

(B) only (S2) is correct

(C) only (S1) is correct

(D) both are correct

कथनों

(S1) : समुच्चय $\{z \in \mathbb{C} - \{-i\} : |z| = 1 \text{ और } \frac{z-i}{z+i} \text{ शुद्धतः वास्तविक हैं}\}$ में तथ्यतः दो अवयव हैं, और

(S2) : समुच्चय $\{z \in \mathbb{C} - \{-1\} : |z| = 1 \text{ और } \frac{z-1}{z+1} \text{ शुद्धतः काल्पनिक हैं}\}$ में अनन्त अवयव हैं

में से

(A) दोनों गलत हैं

(B) केवल (S2) सही है

(C) केवल (S1) सही है

(D) दोनों सही हैं

Ans. B

Sol. S1: $\frac{z-i}{z+i} = \frac{\bar{z}+i}{\bar{z}-i}$

$\Rightarrow z + \bar{z} = 0$

So

S1 is false

S2: $\frac{z-1}{z+1} + \frac{\bar{z}-1}{\bar{z}+1} = 0$

$\Rightarrow \frac{z-1}{z+1} + \frac{\frac{1}{z}-1}{\frac{1}{z}+1} = 0 \Rightarrow \frac{z-1}{z+1} + \frac{1-z}{1+z} = 0$

$\Rightarrow 0 = 0$

So

S2 is correct.

313. If the locus of $z \in \mathbb{C}$, such that $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) + \operatorname{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$, is a circle of radius r and center (a, b) , then $\frac{15ab}{r^2}$ is equal to :

यदि $z \in \mathbb{C}$, जिसके लिए $\operatorname{Re}\left(\frac{z-1}{2z+i}\right) + \operatorname{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$ है, का बिन्दुपथ r त्रिज्या और केन्द्र (a, b) का एक वृत्त है, तो $\frac{15ab}{r^2}$

बराबर है :

(A) 12

(B) 24

(C) 18

(D) 16

Ans. C

Sol. Let $\frac{z-1}{2z+i} = c$

$$\operatorname{Re}(c) + \operatorname{Re}(\bar{c}) = 2\operatorname{Re}(c)$$

put $z = x + iy$

$$c = \frac{x+iy-1}{2x+2iy+i} = \frac{(x-1)+iy}{2x+i(2y+1)}$$

$$= \frac{((x-1)+iy)(2x-i(2y+1))}{4x^2+(2y+1)^2}$$

$$\Rightarrow \operatorname{Re}(c) = \frac{2x(x-1)+y(2y+1)}{4x^2+(2y+1)^2}$$

given $2\operatorname{Re}(c) = 2$

$$\Rightarrow \frac{2x^2-2x+2y^2+y}{4x^2+4y^2+4y+1} = 1$$

$$\Rightarrow x^2+y^2+x+\frac{3y}{2}+\frac{1}{2} = 0$$

$$\text{Centre} \left(-\frac{1}{2}, -\frac{3}{4} \right) \text{ and } r^2 = \frac{1}{4} + \frac{9}{16} - \frac{1}{2} = \frac{5}{16}$$

$$\Rightarrow \frac{15ab}{r^2} = 15 \times \frac{1}{2} \times \frac{3}{4} \times \frac{16}{5} = 18$$

314. Let α be a solution of $x^2 + x + 1 = 0$, and for some a and $b \in \mathbb{R}$, $\begin{bmatrix} 4 & a & b \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$. If

$\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3$, then $m + n$ is equal to :

माना समीकरण $x^2 + x + 1 = 0$ का एक हल α है और किसी $a, b \in \mathbb{R}$ के लिए,

$$\begin{bmatrix} 4 & a & b \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \text{ है। यदि } \frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3, \text{ तो } m + n \text{ बराबर है :}$$

(A) 8

(B) 7

(C) 3

(D) 11

Ans. D

Sol. $4 - a - 2b = 0, 64 - a - 14b = 0, 52 + 2a - 8b = 0.$

$$\Rightarrow a = -6, b = 5$$

Now,

$$\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3 \quad \Rightarrow \quad \frac{4}{\omega^4} + \frac{m}{\omega^{-6}} + \frac{n}{\omega^5} = 3$$

$$4\omega^2 + m + n\omega = 3.$$

$$\Rightarrow \left(\frac{m-3}{4} \right) + \left(\frac{n}{4} \right) \omega + \omega^2 = 0$$

$$\text{Clearly, } \frac{m-3}{4} = 1, \quad \frac{n}{4} = 1 \quad (\because 1 + \omega + \omega^2 = 0)$$

$$m = 7, n = 4$$

315. Let $A = \left\{ \theta \in [0, 2\pi] : 1 + 10 \operatorname{Re} \left(\frac{2 \cos \theta + i \sin \theta}{\cos \theta - 3i \sin \theta} \right) = 0 \right\}$. Then $\sum_{\theta \in A} \theta^2$ is equal to :

माना $A = \left\{ \theta \in [0, 2\pi] : 1 + 10 \operatorname{Re} \left(\frac{2 \cos \theta + i \sin \theta}{\cos \theta - 3i \sin \theta} \right) = 0 \right\}$ है। तो $\sum_{\theta \in A} \theta^2$ बराबर है :

- (A) $6\pi^2$ (B) $8\pi^2$ (C) $\frac{27}{4}\pi^2$ (D) $\frac{21}{4}\pi^2$

Ans. D

Sol. Real part of $\frac{2 \cos \theta + i \sin \theta}{\cos \theta - 3i \sin \theta} = \frac{2 \cos^2 \theta - 3 \sin^2 \theta}{\cos^2 \theta + 9 \sin^2 \theta}$

$$\text{Now, } 1 + 10 \left(\frac{2 \cos^2 \theta - 3 \sin^2 \theta}{\cos^2 \theta + 9 \sin^2 \theta} \right) = 0$$

$$\cos^2 \theta + 9 \sin^2 \theta + 20 \cos^2 \theta - 30 \sin^2 \theta = 0$$

$$\tan^2 \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\sum_{\theta \in A} \theta^2 = \frac{\pi^2}{16} + \frac{9\pi^2}{16} + \frac{25\pi^2}{16} + \frac{49\pi^2}{16} = \frac{21\pi^2}{4}$$

316. Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1) = \frac{-\pi}{4}$, $\arg(z_2) = 0$ and $\arg(z_3) = \frac{\pi}{4}$.

If $|z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1|^2 = \alpha + \beta \sqrt{2}$, $\alpha, \beta \in \mathbb{Z}$, then the value of $\alpha^2 + \beta^2$ is :

माना कि z_1, z_2 और z_3 , $\arg(z_1) = \frac{-\pi}{4}$, $\arg(z_2) = 0$ और $\arg(z_3) = \frac{\pi}{4}$ के साथ वृत्त $|z| = 1$ पर तीन सम्मिश्र संख्याएँ हैं। यदि

$$|z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1|^2 = \alpha + \beta\sqrt{2}, \alpha, \beta \in \mathbb{Z} \text{ है, तो } \alpha^2 + \beta^2 \text{ का मान है :}$$

(A) 31

(B) 24

(C) 29

(D) 41

Ans. C

Sol. $Z_1 = e^{-i\frac{\pi}{4}}, Z_2 = 1, Z_3 = e^{i\frac{\pi}{4}}$

$$|z_1 \bar{z}_2 + z_2 \bar{z}_3 + z_3 \bar{z}_1|^2 = \left| e^{-i\frac{\pi}{4}} \times 1 + 1 \times e^{-i\frac{\pi}{4}} + e^{i\frac{\pi}{4}} \times e^{i\frac{\pi}{4}} \right|^2$$

$$\left| e^{-i\frac{\pi}{4}} + e^{-i\frac{\pi}{4}} + e^{i\frac{\pi}{4}} \right|^2$$

$$= \left| 2e^{-i\frac{\pi}{4}} + i \right|^2 = \left| \sqrt{2} - \sqrt{2}i + i \right|^2$$

$$= (\sqrt{2})^2 + (1 - \sqrt{2})^2 = 2 + 1 + 2 - 2\sqrt{2} = 5 - 2\sqrt{2}$$

$$\alpha = 5, \beta = 2$$

$$\Rightarrow \alpha^2 + \beta^2 = 29$$

317. Let the curve $z(1+i) + \bar{z}(1-i) = 4$, $z \in \mathbb{C}$, divide the region $|z-3| \leq 1$ into two parts of areas α and β . Then $|\alpha - \beta|$ is equal to :

माना कि वक्र $z(1+i) + \bar{z}(1-i) = 4$, $z \in \mathbb{C}$, क्षेत्र $|z-3| \leq 1$ को क्षेत्रफलों α और β के दो भागों में विभाजित करता है। तब $|\alpha - \beta|$ बराबर है :

(A) $1 + \frac{\pi}{2}$

(B) $1 + \frac{\pi}{6}$

(C) $1 + \frac{\pi}{3}$

(D) $1 + \frac{\pi}{4}$

Ans. A

Sol. $z(1+i) + \bar{z}(1-i) = 4$

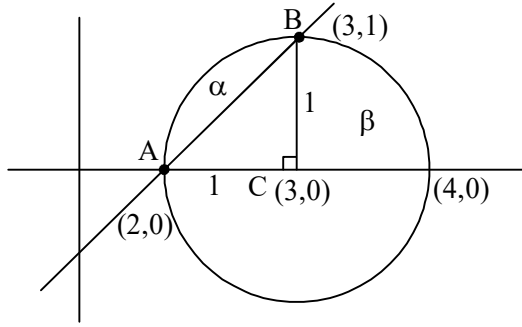
put $x = x + iy$

$$(x+iy)(1+i) + (x-iy)(1-i) = 4$$

$$x-y+iy+ix+x-ix-iy-y=4$$

$$x-y=2$$

$$|Z-3| \leq 1 \quad \text{circle}$$



$$\alpha + \beta = \pi r^2 = \pi(1)^2$$

$$\alpha + \beta = \pi$$

Now $\alpha = \text{Area of sector} - \text{Area of } \triangle ABC$

$$= \frac{\pi(1)^2}{4} - \frac{1}{2} \times 1 \times 1$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

$$\text{So, } \beta = \pi - \alpha$$

$$|\alpha - \beta| = |2\alpha - \pi|$$

$$= \left| \frac{\pi}{2} - 1 - \pi \right|$$

$$= \left| -1 - \frac{\pi}{2} \right| = 1 + \frac{\pi}{2}$$

318. Let $\left| \frac{\bar{z} - i}{2\bar{z} + i} \right| = \frac{1}{3}$, $z \in \mathbb{C}$, be the equation of a circle with center at C. If the area of the triangle, whose vertices are at the points (0,0), C and $(\alpha, 0)$ is 11 square units, then α^2 equals :

माना कि $\left| \frac{\bar{z} - i}{2\bar{z} + i} \right| = \frac{1}{3}$, $z \in \mathbb{C}$, केन्द्र C के साथ एक वृत्त का समीकरण है। यदि उस त्रिभुज, जिसके शीर्ष, बिन्दुओं (0,0), C और

$(\alpha, 0)$ पर हैं, का क्षेत्रफल 11 वर्ग इकाई है, तो α^2 बराबर है :

- (A) 50 (B) $\frac{121}{25}$ (C) 100 (D) $\frac{81}{25}$

Ans. C

Sol. $3|x - iy - i| = |2(x - iy) + i|$

$$3|x - i(y + 1)| = |2x + i(1 - 2y)|$$

$$3\sqrt{x^2 + (y + 1)^2} = \sqrt{4x^2 + (1 - 2y)^2}$$

$$9x^2 + 9y^2 + 18y + 9 = 4x^2 + 4y^2 - 4y + 1$$

$$5x^2 + 5y^2 + 22y + 8 = 0$$

$$x^2 + y^2 + \frac{22}{5}y + \frac{8}{5} = 0$$

$$C\left(0, -\frac{11}{5}\right)$$

$$\frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 0 & -\frac{11}{5} & 1 \\ \alpha & 0 & 1 \end{vmatrix} = 11$$

$$\left| \frac{11\alpha}{5} \right| = 22$$

$$\left| \frac{\alpha}{5} \right| = 2$$

$$\alpha^2 = 100$$

319. The number of complex numbers z , satisfying $|z| = 1$ and $\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$, is :

$|z| = 1$ और $\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$ को संतुष्ट करने वाली सम्मिश्र संख्याओं z की संख्या है :

- (A) 6 (B) 4 (C) 10 (D) 8

Ans. D

Sol. $|z| = 1$ & $\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$

Let $z = e^{i\theta}$

$$\bar{z} = e^{-i\theta}$$

$$\therefore \frac{z}{\bar{z}} = \frac{e^{i\theta}}{e^{-i\theta}} = e^{2i\theta}$$

$$|e^{2i\theta} + e^{-i2\theta}| = 1$$

$$|\cos 2\theta + i \sin 2\theta + \cos 2\theta - i \sin 2\theta| = 1$$

$$|\cos 2\theta| = \frac{1}{2}$$

$$\cos 2\theta = \pm \frac{1}{2}$$

Number of solution $\equiv 8$

320. If α and β are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then

$$16 \cdot \operatorname{Re} \left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} \right) \cdot \operatorname{Im} \left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} \right) \text{ is equal to :}$$

यदि α और β , समीकरण $2z^2 - 3z - 2i = 0$, जहाँ $i = \sqrt{-1}$ है, के मूल हैं, तो

$$16 \cdot \operatorname{Re} \left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} \right) \cdot \operatorname{Im} \left(\frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} \right) \text{ बराबर है :}$$

(A) 398

(B) 409

(C) 441

(D) 312

Ans. C

Sol. α is a root of $2z^2 - 3z - 2i = 0$

$$\therefore 2\alpha^2 - 3\alpha - 2i = 0$$

$$\Rightarrow 2\alpha^2 - 2i = 3\alpha$$

$$\Rightarrow \alpha - \frac{i}{\alpha} = \frac{3}{2}$$

$$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} - 2i = \frac{9}{4}$$

$$\Rightarrow \alpha^2 - \frac{1}{\alpha^2} = \frac{9}{4} + 2i$$

$$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} - 2 = \frac{81}{16} - 4 + 9i$$

$$\Rightarrow \alpha^4 + \frac{1}{\alpha^4} = \frac{81-32}{16} + 9i = \frac{49}{16} + 9i$$

$$\Rightarrow \text{Similarly } \beta^4 + \frac{1}{\beta^4} = \frac{49}{16} + 9i$$

$$\text{Now, } \frac{\alpha^{19} + \beta^{19} + \alpha^{11} + \beta^{11}}{\alpha^{15} + \beta^{15}} = \frac{\alpha^{15} \left(\alpha^4 + \frac{1}{\alpha^4} \right) + \beta^{15} \left(\beta^4 + \frac{1}{\beta^4} \right)}{\alpha^{15} + \beta^{15}}$$

$$= \frac{\left(\frac{49}{16} + 9i \right) (\alpha^{15} + \beta^{15})}{\alpha^{15} + \beta^{15}} = \frac{49}{16} + 9i$$

$$\therefore \text{Ans.} = 16 \times \frac{49}{16} \times 9 = 441$$